Fair Test Practical · Pupil Resources

Learning goal: I can record repeated results, spot an odd reading, and use the data to answer our question.

Arrival Questions (In your book)

1. What upward force slows a falling parachute?

2. In Box 1 (2.1 s, 2.0 s, 2.2 s), are the repeats close together?

3. What do we call an odd reading like 0.8 s?

4. Can you ask an adult to help?

Starter Question

Why did the 30x30 cm parachute take 3.4 s to fall compared with 1.2 s for the 10x10 cm parachute?

A. Its larger surface area met more air particles, producing greater air resistance to slow the fall.

B. The 10 g load lost mass in the air.

C. Gravity stopped acting on the large parachute.

Method Strip

1. Check the 10 g mass is attached and hold the 10x10 cm canopy level with the 1.50 m mark.

2. Use '3, 2, 1, drop' to release without pushing and time the fall in seconds. STOP AND CHECK.

3. Record three trials for Small, Medium, and Large canopies (or inspect the supplied table).

4. Circle any anomalous reading and write the typical time for each size.

5. Write your 3-part conclusion in your book.

Exit Ticket

Which parachute took the longest time to fall: 10x10 cm (1.2 s) or 30x30 cm (3.4 s)?

Why does a larger parachute canopy fall more slowly?

What should you do if one repeat trial is an anomaly because the strings tangled?